Execution Pipeline
Step 1. Predict Goal Pixel
(u, v)
Back-project 2D pixel to 3D waypoint
Step 2. Project 3D Waypoint
(x, y, z)
Convert the waypoint into low-level actions via a local planner
Step 3. Low-Level Action Execution
Forward
Stop
Right
Left
Vision Model Backbone
Padded Current Observation Image
History Images (ViKeyMem)
Vision Encoder
Projection Layer
Current Image Padding Token + $e_{padding}$
Current Image RGB token
History Trajectory Token + $e_{trajectory}$
History RGB token
Large Language Model Backbone
You are an indoor navigation agent following...... <Task Introduction>
Instruction: <Current Episode Instruction>
Input:
- Current Step egocentric RGB image ,where the left/right/bottom gray padding indicates left/right turns and stop.
- History Images: Previous <N> egocentric RGB keyframes (latest), , (oldest).
Output: a pixel coordinate pair in the format XXX, YYY\n"
Vision-Language Model (InternVL3.0)
Training Objectives:
$\mathcal{L} = \mathcal{L}_{CE} + \lambda\mathcal{L}_{Numeric} + \mathcal{L}_{Angular}$
X_1 X_2 X_3 , Y_1 Y_2 Y_3
Soft Predicted Pixel Coordinate
$\hat{u}$
$\hat{v}$
$\hat{u}_{gt}, \hat{v}_{gt}$
ℓ_1
$\hat{u}, \hat{v}$
$\Delta\theta$
$\mathcal{L}_{CE} = -\sum_{t-1}^{T} \log p(y_t|y_{<t}, x)$
$\mathcal{L}_{Numeric} = ||(\hat{u}, \hat{v}) - (u_{gt}, v_{gt})||_1$
$\mathcal{L}_{Angular} = 1 - \cos(\Delta\theta)$